25901339 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCIAM200 - PRINCIPLES OF ROBOTICS AND AUTOMATION

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. The driver circuit in stepper motor is used to [K] / [CO1]

   a). Reduce heat b). Amplify control signals c). Change color d). Store energy

2. ASIMO robot was developed by [U] / [CO1]

   a). Toyota b). Honda c). Tesla d). Siemens

3. Which sensor works based on reflection of emitted IR light? [U] / [CO1]

   a). PIR sensor b). IR proximity sensor c). Piezoelectric sensor d). Thermistor

4. The stator of a stepper motor contains [U] / [CO1]

   a). Permanent magnets only b). Windings c). Bearings d). Brushes

5. If the IR LED fails in an IR proximity sensor, the sensor [U] / [CO1]

a). Still works normally b). Cannot detect object c). Detects temperature d). Produces heat

6. PIR sensor has usually [U] / [CO2]

a). 2 pins b). 3 pins c). 4 pins d). 5 pins

7. Which device is commonly used between PLC and motor? [U] / [CO2]

a). LED b). Relay c). Sensor d). Timer

8. PLC stands for [U] / [CO2]

a). Programmable Logic Control b). Programmable Logic Controller

c). Power Logic Controller d). Process Logic Circuit

9. Actuators in robotics are used to [K] / [CO3]

a). Sense environmental changes b). Process information

c). Convert control signals into motion d). Store feedback

10. Which PLC instruction is used to control weld time? [U] / [CO3]

a). Counter b). Timer c). Comparator d). Latch

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Define a PIR sensor. Explain why it is called a passive sensor. [U] / [CO1]

12. Write a short note on piezoelectric materials with examples. [U] / [CO1]

13. Define a servo motor and explain its basic purpose. [U] / [CO1]

14. Explain types of I/O modules in PLC. [U] / [CO2]

15. Explain how a retentive timer maintains accumulated value during power loss and how it is reset. [U] / [CO2]

16. Draw a ladder logic to stop the conveyor motor using an emergency stop push button. [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Discuss the purpose of industrial automation and its role in improving productivity, safety, and quality. [U] / [CO1]

18. Describe sourcing and sinking output modules in PLC. [U] / [CO2]

19. Define pick-and-place robots and explain their working principle with a neat labeled diagram. [K] / [CO3]

20. What is the purpose of a controller in a water level control system? [K] / [CO3]

[10x2 = 20]